

**LAPORAN PRAKTIKUM
APLIKASI BERBASIS PLATFORM**

**PERTEMUAN 12
MOBILE**



Disusun oleh:
MUHAMMAD ARSYAD ZAIDAN
2311102058
S1 IF-11-04

Dosen Pengampu:
Cahyo Prihantoro, S.Kom., M.Eng

PROGRAM STUDI S1 INFORMATIKA
FAKULTAS INFORMATIKA
TELKOM UNIVERSITY PURWOKERTO
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LINK GITHUB :

1. Main.dart

Source code :

```
import 'dart:io';
import 'package:flutter/material.dart';
import 'package:image_picker/image_picker.dart';
import
'package:flutter_local_notifications/flutter_local_noti
fications.dart';

final FlutterLocalNotificationsPlugin
flutterLocalNotificationsPlugin =
    FlutterLocalNotificationsPlugin();

void main() {
    runApp(const MyApp());
}

class MyApp extends StatelessWidget {
    const MyApp({super.key});

    @override
    Widget build(BuildContext context) {
        return MaterialApp(
            debugShowCheckedModeBanner: false,
            title: 'Notifikasi & API Perangkat Keras',
            theme: ThemeData(primarySwatch: Colors.blue),
            home: const HomePage(),
        );
    }
}

class HomePage extends StatefulWidget {
    const HomePage({super.key});

    @override
    State<HomePage> createState() => _HomePageState();
}

class _HomePageState extends State<HomePage> {
    File? _image;
    final ImagePicker _picker = ImagePicker();

    @override
    void initState() {
        super.initState();
        _initializeNotifications();
    }

    Future<void> _initializeNotifications() async {
        const AndroidInitializationSettings androidSettings
        =
            AndroidInitializationSettings('@mipmap/ic_launc
her');
    }
```

```

        const InitializationSettings settings =
        InitializationSettings(
            android: androidSettings,
        );

        await
flutterLocalNotificationsPlugin.initialize(settings);

        await flutterLocalNotificationsPlugin
            .resolvePlatformSpecificImplementation<
                AndroidFlutterLocalNotificationsPlugin
            >()
            ?.requestNotificationsPermission();
    }

    Future<void> _showNotification(String message) async
    {
        const AndroidNotificationDetails androidDetails =
        AndroidNotificationDetails(
            'photo_channel',
            'Photo Notifications',
            channelDescription:
                'Notifikasi saat foto berhasil dipilih
atau diambil',
            importance: Importance.max,
            priority: Priority.high,
        );

        const NotificationDetails notificationDetails =
        NotificationDetails(
            android: androidDetails,
        );

        await flutterLocalNotificationsPlugin.show(
            DateTime.now().millisecond,
            'Praktikum Flutter',
            message,
            notificationDetails,
        );
    }

    Future<void> _pickFromCamera() async {
        final XFile? photo = await
        _picker.pickImage(source: ImageSource.camera);

        if (photo != null) {
            setState(() {
                _image = File(photo.path);
            });

            await _showNotification('Foto berhasil diambil
dari kamera');
        }
    }

```

```

Future<void> _pickFromGallery() async {
  final XFile? photo = await
_picker.pickImage(source: ImageSource.gallery);

  if (photo != null) {
    setState(() {
      _image = File(photo.path);
    });

    await _showNotification('Foto berhasil dipilih
dari galeri');
  }
}

@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(
      title: Column(
        mainAxisAlignment: MainAxisAlignment.center,
        children: [
          const Text(
            'Muhammad Arsyad Zaidan',
            style: TextStyle(fontSize: 20,
fontWeight: FontWeight.bold),
          ),
          const Text(
            'Tambah foto dari camera/galeri',
            style: TextStyle(fontSize: 14,
fontWeight: FontWeight.normal),
          ),
        ],
      ),
      centerTitle: true,
      toolbarHeight: 70,
    ),
    body: Padding(
      padding: const EdgeInsets.all(20),
      child: Column(
        children: [
          Expanded(
            child: Center(
              child: _image == null
                ? const Text(
                    'Belum ada foto dipilih',
                    style: TextStyle(fontSize: 18),
                )
                : Image.file(_image!, fit:
BoxFit.contain),
            ),
          ),
          const SizedBox(height: 20),
          SizedBox(
            width: double.infinity,
            child: ElevatedButton.icon(

```

```

        onPressed: _pickFromCamera,
        icon: const Icon(Icons.camera_alt),
        label: const Text('Buka Kamera'),
      ),
    ),
    const SizedBox(height: 15),
    SizedBox(
      width: double.infinity,
      child: ElevatedButton.icon(
        onPressed: _pickFromGallery,
        icon: const Icon(Icons.photo_library),
        label: const Text('Pilih dari Galeri'),
      ),
    ),
  ],
),
);
}
}

```

Penjelasan :

Aplikasi yang dirancang secara khusus untuk mengintegrasikan fitur pengambilan gambar serta akses galeri dengan sistem pemberitahuan langsung pada perangkat Android. Pada saat aplikasi pertama kali dijalankan oleh pengguna, sistem secara otomatis memicu dialog permintaan izin (permission requests) guna mengaktifkan fungsionalitas notifikasi lokal serta akses perangkat keras kamera.

Di dalam antarmuka utama aplikasi, yang mengusung identitas personal pada bagian header, disediakan dua tombol interaktif sebagai kendali utama operasional. Tombol pertama berfungsi untuk menginisialisasi kamera bawaan perangkat agar pengguna dapat menangkap momen secara langsung, sedangkan tombol kedua dialokasikan untuk membuka pustaka galeri guna memilih gambar yang telah tersimpan sebelumnya.

Alur kerja aplikasi berjalan secara simultan; segera setelah pengguna menyelesaikan proses pengambilan atau pemilihan gambar, sistem akan mengeksekusi dua tindakan sekaligus. Pertama, berkas foto yang berhasil didapatkan akan langsung dimuat dan ditampilkan di area tengah layar sebagai pratinjau (preview) adaptif. Kedua, aplikasi akan memicu framework notifikasi lokal untuk memunculkan pesan pemberitahuan instan (banner notification) di bagian atas layar handphone, yang berfungsi sebagai konfirmasi formal bahwa tindakan penambahan foto telah berhasil diselesaikan dengan aman.

2. AndroidManifest.xml

Source code :

```

<manifest
xmlns:android="http://schemas.android.com/apk/res/android">
  <uses-permission
android:name="android.permission.CAMERA"/>
  <uses-permission
android:name="android.permission.POST_NOTIFICATIONS"/>

```

```

<application
    android:label="NotifAplikasi_2311102058"
    android:name="${applicationName}"
    android:icon="@mipmap/ic_launcher">

    <activity
        android:name=".MainActivity"
        android:exported="true"
        android:launchMode="singleTop"
        android:taskAffinity=""
        android:theme="@style/LaunchTheme"
        android:configChanges="orientation|keyboardHidden
|keyboard|screenSize|smallestScreenSize|locale|layoutDirectio
n|fontScale|screenLayout|density|uiMode"
        android:hardwareAccelerated="true"
        android:windowSoftInputMode="adjustResize">

        <meta-data
            android:name="io.flutter.embedding.android.Norm
alTheme"
            android:resource="@style/NormalTheme"
        />

        <intent-filter>
            <action
                android:name="android.intent.action.MAIN"/>
            <category
                android:name="android.intent.category.LAUNCHER"/>
            </intent-filter>
        </activity>

        <receiver
            android:name="com.dexterous.flutterlocalnotifications.Schedul
edNotificationReceiver" android:exported="false" />
        <receiver
            android:name="com.dexterous.flutterlocalnotifications.Schedul
edNotificationBootReceiver" android:exported="false">
            <intent-filter>
                <action
                    android:name="android.intent.action.BOOT_COMPLETED"/>
                <action
                    android:name="android.intent.action.MY_PACKAGE_REPLACED"/>
                <action
                    android:name="android.quickboot.POWERON"/>
                <action
                    android:name="com.htc.intent.action.QUICKBOOT_POWERON"/>
            </intent-filter>
        </receiver>

        <meta-data
            android:name="flutterEmbedding"
            android:value="2" />
    </application>

```

```

<queries>
  <intent>
    <action
android:name="android.intent.action.PROCESS_TEXT"/>
    <data android:mimeType="text/plain"/>
  </intent>
</queries>
</manifest>

```

Penjelasan :

Dokumen konfigurasi AndroidManifest.xml ini berfungsi untuk mendaftarkan izin akses perangkat keras kamera dan otorisasi pengiriman notifikasi pada sistem operasi Android. Di dalamnya, nama aplikasi diatur melalui atribut label sekaligus menetapkan MainActivity sebagai titik masuk utama saat aplikasi dijalankan oleh pengguna. Selain itu, file ini mendaftarkan komponen penerima pesan (receiver) agar sistem notifikasi lokal tetap dapat berfungsi dengan stabil dan responsif di latar belakang perangkat.

3. Pubspec.yaml

Source code :

```

name: Pertemuan12_2311102058
description: "A new Flutter project."
# The following line prevents the package from being
# accidentally published to
# pub.dev using `flutter pub publish`. This is
# preferred for private packages.
publish_to: 'none' # Remove this line if you wish to
# publish to pub.dev

# The following defines the version and build number
# for your application.
# A version number is three numbers separated by dots,
# like 1.2.43
# followed by an optional build number separated by a
# +.
# Both the version and the builder number may be
# overridden in flutter
# build by specifying --build-name and --build-number,
# respectively.
# In Android, build-name is used as versionName while
# build-number used as versionCode.
# Read more about Android versioning at
# https://developer.android.com/studio/publish/versioning
# In iOS, build-name is used as
# CFBundleShortVersionString while build-number is used
# as CFBundleVersion.
# Read more about iOS versioning at
#
# https://developer.apple.com/library/archive/documentati
# on/General/Reference/InfoPlistKeyReference/Articles/Cor
# eFoundationKeys.html
# In Windows, build-name is used as the major, minor,
# and patch parts

```



```
# of the product and file versions while build-number
is used as the build suffix.
version: 1.0.0+1

environment:
  sdk: ^3.11.5

# Dependencies specify other packages that your package
needs in order to work.
# To automatically upgrade your package dependencies to
the latest versions
# consider running `flutter pub upgrade --major-
versions`. Alternatively,
# dependencies can be manually updated by changing the
version numbers below to
# the latest version available on pub.dev. To see which
dependencies have newer
# versions available, run `flutter pub outdated`.
dependencies:
  flutter:
    sdk: flutter
  image_picker: ^1.1.2
  flutter_local_notifications: ^17.2.1

  # The following adds the Cupertino Icons font to your
application.
  # Use with the CupertinoIcons class for iOS style
icons.
  cupertino_icons: ^1.0.8

dev_dependencies:
  flutter_test:
    sdk: flutter

  # The "flutter_lints" package below contains a set of
recommended lints to
  # encourage good coding practices. The lint set
provided by the package is
  # activated in the `analysis_options.yaml` file
located at the root of your
  # package. See that file for information about
deactivating specific lint
  # rules and activating additional ones.
  flutter_lints: ^6.0.0

# For information on the generic Dart part of this
file, see the
# following page: https://dart.dev/tools/pub/pubspec

# The following section is specific to Flutter
packages.
flutter:

  # The following line ensures that the Material Icons
font is
```

```

# included with your application, so that you can use
the icons in
# the material Icons class.
uses-material-design: true

# To add assets to your application, add an assets
section, like this:
# assets:
#   - images/a_dot_burr.jpeg
#   - images/a_dot_ham.jpeg

# An image asset can refer to one or more resolution-
specific "variants", see
# https://flutter.dev/to/resolution-aware-images

# For details regarding adding assets from package
dependencies, see
# https://flutter.dev/to/asset-from-package

# To add custom fonts to your application, add a
fonts section here,
# in this "flutter" section. Each entry in this list
should have a
# "family" key with the font family name, and a
"fonts" key with a
# list giving the asset and other descriptors for the
font. For
# example:
# fonts:
#   - family: Schyler
#     fonts:
#       - asset: fonts/Schyler-Regular.ttf
#       - asset: fonts/Schyler-Italic.ttf
#         style: italic
#   - family: Trajan Pro
#     fonts:
#       - asset: fonts/TrajanPro.ttf
#       - asset: fonts/TrajanPro_Bold.ttf
#         weight: 700
#
# For details regarding fonts from package
dependencies,
# see https://flutter.dev/to/font-from-package

```

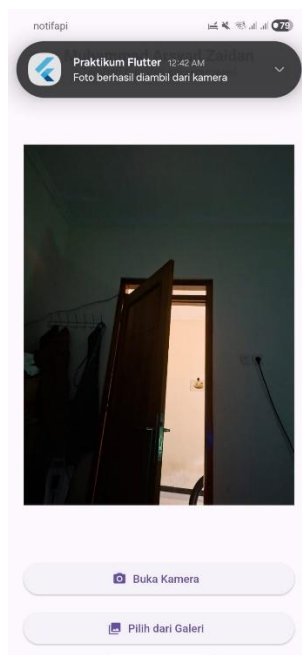
Penjelasan :

File konfigurasi pubspec.yaml ini berfungsi untuk mendaftarkan dependensi atau pustaka pihak ketiga yang diperlukan dalam pengembangan proyek Flutter. Package image_picker ditambahkan agar aplikasi memiliki kemampuan untuk mengakses perangkat keras kamera serta membuka galeri foto lokal pada perangkat. Sementara itu, package flutter_local_notifications disematkan guna menyediakan infrastruktur yang diperlukan untuk membuat dan menampilkan sistem pemberitahuan langsung di layar HP pengguna.

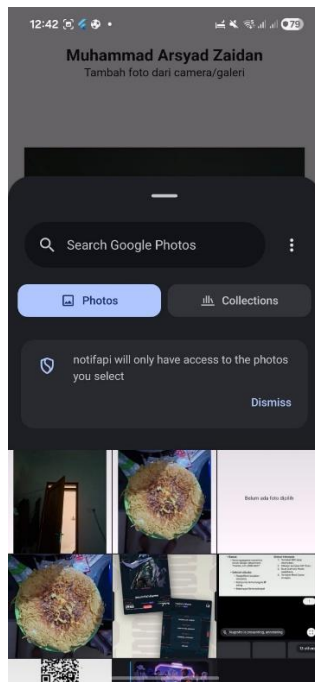
- Output code :
- 1. Home
- Halaman yang menunjukkan fitur ambil foto dan galeri (kondisi belum ambil foto).



- 2. Camera
- Berikut adalah foto dari camera.



- 3. Galleri
- Sebelum mengambil foto dari galeri, diperlukan izin dari android. Pilih allow saja dan foto lalu foto akan ter-upload.



- Berikut kondisi ketika foto galeri sudah selesai di-upload

